

Generating Functionals for ϕ^4 Theory

Perimeter Scholars International, Quantum Field Theory II, Homework 1; 10 worked parts.

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Quantum Field Theory

Question prompts paraphrased from the official Perimeter PSI Online sheet; the equations and notation follow the source.

1. Computing Correlation Functions Using Generating Functionals

Work in Euclidean signature with the interacting scalar-field action

$$S[\phi] = S_0[\phi] + \int d^4x V(\phi), \quad V(\phi) = \frac{g}{4!} \phi^4, \quad (1.1)$$

where $S_0[\phi]$ is the free Euclidean action.

a) Introduce a source $J(x)$ and consider the free generating functional

$$\mathcal{Z}_0[J] = \int \mathcal{D}\phi \exp \left[-\frac{1}{\hbar} \left(S_0[\phi] - \int d^4x J(x)\phi(x) \right) \right]. \quad (1.2)$$

Evaluate this Gaussian functional integral. As a hint, first shift the integration variable by a function determined by J .

Solution. In the Euclidean picture, we have

$$S_0[\phi] = -\frac{1}{2} \phi (\partial^2 - m^2) \phi. \quad (1.3)$$

To evaluate the integral, we make the shift of the integration variable,

$$\bar{\phi}(x) = \phi(x) + \int d^d y \Delta(x-y) J(y) \equiv \phi_x + \Delta_{xy} J_y, \quad (1.4)$$

where we have used the generalised summation convention and $(\partial^2 - m^2)\Delta(x-y) = \delta^d(x-y)$. Since $\mathcal{D}\phi = \mathcal{D}\bar{\phi}$, we have

$$\begin{aligned} \mathcal{Z}_0[J] &= \int \mathcal{D}\bar{\phi} \exp \left[-\frac{1}{\hbar} \left(-\frac{1}{2} \bar{\phi}_x (\partial^2 - m^2)_{xy} \bar{\phi}_y + \frac{1}{2} J_x \Delta_{xy} J_y \right) \right] \\ &= \exp \left[-\frac{1}{\hbar} \int d^d x \int d^d y \frac{1}{2} J(x) \Delta(x-y) J(y) \right] \mathcal{Z}_0[0]. \end{aligned} \quad (1.5)$$

b) For an arbitrary polynomial interaction $V(\phi)$, establish the identity

$$\begin{aligned} &\exp \left[-\frac{1}{\hbar} \int d^4x V(\phi) + \frac{1}{\hbar} \int d^4x J(x)\phi(x) \right] \\ &= \exp \left[-\frac{1}{\hbar} \int d^4x V \left(\hbar \frac{\delta}{\delta J(x)} \right) \right] \exp \left[\frac{1}{\hbar} \int d^4x J(x)\phi(x) \right]. \end{aligned} \quad (1.6)$$

Solution.

$$\begin{aligned} &\exp \left[-\frac{1}{\hbar} \int d^4x V(\phi) + \frac{1}{\hbar} \int d^4x J(x)\phi(x) \right] \\ &= \sum_{n=0}^{\infty} \frac{1}{n!} \left(-\frac{1}{\hbar} \right)^n \left(\int d^4x V(\phi) \right)^n \exp \left(\frac{1}{\hbar} \int d^4x J(x)\phi(x) \right) \\ &= \sum_{n=0}^{\infty} \frac{1}{n!} \left(-\frac{1}{\hbar} \right)^n \left(\int d^4x V \left(\hbar \frac{\delta}{\delta J(x)} \right) \right)^n \exp \left(\frac{1}{\hbar} \int d^4x J(x)\phi(x) \right) \\ &= \exp \left[-\frac{1}{\hbar} \int d^4x V \left(\hbar \frac{\delta}{\delta J(x)} \right) \right] \exp \left[\frac{1}{\hbar} \int d^4x J(x)\phi(x) \right]. \end{aligned} \quad (1.7)$$

c) For the interacting theory, define

$$\mathcal{Z}[J] = \int \mathcal{D}\phi \exp \left[-\frac{1}{\hbar} \left(S[\phi] - \int d^4x J(x)\phi(x) \right) \right]. \quad (1.8)$$

Use the preceding result to express $\mathcal{Z}[J]$ formally as a functional differential operator acting on $\mathcal{Z}_0[J]$.

Solution. From b), we have

$$Z[J] = \int \mathcal{D}\phi \exp \left[-\frac{1}{\hbar} \int d^4x V \left(\hbar \frac{\delta}{\delta J(x)} \right) \right] \exp \left[-\frac{1}{\hbar} \left(S_0[\phi] - \int d^4x J(x)\phi(x) \right) \right]. \quad (1.9)$$

Then, using the same field redefinition in a), we find

$$Z[J] = \exp \left[-\frac{1}{\hbar} \int d^4x V \left(\hbar \frac{\delta}{\delta J(x)} \right) \right] \exp \left[-\frac{1}{\hbar} \int d^4x \int d^4y \frac{1}{2} J(x) \Delta(x-y) J(y) \right] Z_0[0], \quad (1.10)$$

or equivalently,

$$Z[J] = \exp \left[-\frac{1}{\hbar} \int d^4x V \left(\hbar \frac{\delta}{\delta J(x)} \right) \right] Z_0[J]. \quad (1.11)$$

d) The two-point Green function may be obtained from

$$G(x_1, x_2) = \langle \phi(x_1) \phi(x_2) \rangle = \frac{\hbar^2}{\mathcal{Z}[0]} \frac{\delta^2 \mathcal{Z}[J]}{\delta J(x_1) \delta J(x_2)} \Big|_{J=0}. \quad (1.12)$$

Compute its perturbative expansion through order g^2 in ϕ^4 theory. At each order, list every contribution without evaluating the position-space integrals, draw its Feynman diagram, and give the correct symmetry factor.

Solution. Using the results from above, we have

$$G(x_1, x_2) = \hbar^2 \frac{\delta^2}{\delta_1 \delta_2} \exp \left[-\frac{1}{\hbar} \int d^4x V \left(\hbar \frac{\delta}{\delta_x} \right) \right] \exp \left[-\frac{1}{\hbar} \frac{1}{2} J_x \Delta_{xy} J_y \right] \Big|_{J=0} \frac{Z_0[0]}{Z[0]}, \quad (1.13)$$

where $Z[0]$ can be written as

$$Z[0] = \exp \left[-\frac{1}{\hbar} \int d^4x V \left(\hbar \frac{\delta}{\delta_x} \right) \right] \exp \left[-\frac{1}{\hbar} \frac{1}{2} J_x \Delta_{xy} J_y \right] \Big|_{J=0} Z_0[0]. \quad (1.14)$$

Expanding the first exponential to order g^2 , we have

$$\begin{aligned} \exp \left[-\frac{1}{\hbar} \int d^4x V \left(\hbar \frac{\delta}{\delta_x} \right) \right] &= 1 - \frac{1}{\hbar} \frac{g}{4!} \int d^4x \left(\hbar \frac{\delta}{\delta_x} \right)^4 \\ &\quad + \frac{1}{2\hbar^2} \left(\frac{g}{4!} \right)^2 \int d^4x \left(\hbar \frac{\delta}{\delta_x} \right)^4 \int d^4y \left(\hbar \frac{\delta}{\delta_y} \right)^4 + \dots \end{aligned} \quad (1.15)$$

At order g^0 , we simply have $Z[0] = Z_0[0]$, and hence

$$G_{12} = -\hbar \Delta_{12} + \dots \quad (1.16)$$

At order g^1 , we need to evaluate

$$\begin{aligned}
& \left(\hbar \frac{\delta}{\delta_x} \right)^4 \exp \left[-\frac{1}{\hbar} \frac{1}{2} J_x \Delta_{xy} J_y \right] \\
&= \left(\hbar \frac{\delta}{\delta_x} \right)^3 \left(-\Delta_{xy} J_y \exp \left[-\frac{1}{\hbar} \frac{1}{2} J_x \Delta_{xy} J_y \right] \right) \\
&= \left(\hbar \frac{\delta}{\delta_x} \right)^2 \left((-\hbar \Delta_{xx} + \Delta_{xy} \Delta_{xz} J_y J_z) \exp \left[-\frac{1}{\hbar} \frac{1}{2} J_x \Delta_{xy} J_y \right] \right) \\
&= \left(\hbar \frac{\delta}{\delta_x} \right) \left((3\hbar \Delta_{xx} \Delta_{xy} J_y - \Delta_{xy} \Delta_{xz} \Delta_{xw} J_y J_z J_w) \exp \left[-\frac{1}{\hbar} \frac{1}{2} J_x \Delta_{xy} J_y \right] \right) \\
&= \left((3\hbar^2 \Delta_{xx} \Delta_{xx} - 6\hbar \Delta_{xy} \Delta_{xz} \Delta_{xx} J_y J_z + \Delta_{xy} \Delta_{xz} \Delta_{xw} \Delta_{xu} J_y J_z J_w J_u) \exp \left[-\frac{1}{\hbar} \frac{1}{2} J_x \Delta_{xy} J_y \right] \right).
\end{aligned} \tag{1.17}$$

From this, we can read out

$$Z[0] = Z_0[0] \left(1 - \frac{g}{8} \hbar \int d^d x \Delta_{xx}^2 + \dots \right). \tag{1.18}$$

Diagrammatically, the order- g vacuum contribution is

$$\frac{Z[0]}{Z_0[0]} = 1 + \text{diagram of two circles connected by a dot} + \dots, \quad S = 8. \tag{1.19}$$

Hence

$$\begin{aligned}
G_{12} &= \left[-\hbar \Delta_{12} + \frac{g}{8} \hbar^2 \Delta_{12} \int d^d x \Delta_{xx}^2 + \frac{g}{2} \hbar^2 \int d^d x \Delta_{xx} \Delta_{1x} \Delta_{2x} + \mathcal{O}(g^2) \right] \\
&\quad \times \left[1 + \frac{g}{8} \hbar \int d^d y \Delta_{yy}^2 + \mathcal{O}(g^2) \right] \\
&= -\hbar \Delta_{12} + \frac{g}{2} \hbar^2 \int d^d x \Delta_{xx} \Delta_{1x} \Delta_{2x} + \mathcal{O}(g^2).
\end{aligned} \tag{1.20}$$

The disconnected free propagator times the vacuum bubble has cancelled. The surviving diagrams are

$$G_{12} = x_1 \text{ --- } x_2 + x_1 \text{ --- } \text{diagram of a circle with a dot} \text{ --- } x_2 + \mathcal{O}(g^2), \quad S_{g^0} = 1, \quad S_{g^1} = 2. \tag{1.21}$$

Now, we turn to order g^2 , where we need to evaluate

$$\begin{aligned}
& \left(\hbar \frac{\delta}{\delta_x} \right)^4 \left(\hbar \frac{\delta}{\delta_y} \right)^4 \exp \left[-\frac{1}{\hbar} \frac{1}{2} J_x \Delta_{xy} J_y \right] \\
&= \left(\hbar \frac{\delta}{\delta_y} \right)^4 \left[\left(3\hbar^2 \Delta_{xx}^2 - 6\hbar \Delta_{xx} \Delta_{xv} \Delta_{xw} J_v J_w + \Delta_{xv} \Delta_{xw} \Delta_{xz} \Delta_{xu} J_v J_w J_z J_u \right) \exp \left[-\frac{1}{\hbar} \frac{1}{2} J_x \Delta_{xy} J_y \right] \right] \\
&= \left\{ -\hbar^3 \left[18\Delta_{xx} (\Delta_{xx} \Delta_{yy} + 4\Delta_{xy}^2) \Delta_{yr} \Delta_{ys} J_r J_s \right. \right. \\
&\quad + 48\Delta_{xy} (3\Delta_{xx} \Delta_{yy} + 2\Delta_{xy}^2) \Delta_{xv} \Delta_{yr} J_v J_r \\
&\quad + 18\Delta_{yy} (\Delta_{xx} \Delta_{yy} + 4\Delta_{xy}^2) \Delta_{xv} \Delta_{xw} J_v J_w \left. \right] \\
&\quad + \hbar^4 \left(9\Delta_{xx}^2 \Delta_{yy}^2 + 72\Delta_{xx} \Delta_{yy} \Delta_{xy}^2 + 24\Delta_{xy}^4 \right) \left. \right\} \exp \left[-\frac{1}{\hbar} \frac{1}{2} J_x \Delta_{xy} J_y \right] \\
&\quad + \text{terms containing more than two currents.}
\end{aligned} \tag{1.22}$$

We have discarded terms with more than two currents because they vanish after the two external derivatives are taken and J is set to zero. From this, we find

$$Z[0] = Z_0[0] \left[1 - \frac{g}{8} \hbar \int d^d x \Delta_{xx}^2 + g^2 \hbar^2 \int d^d x d^d y \left(\frac{1}{128} \Delta_{xx}^2 \Delta_{yy}^2 + \frac{1}{16} \Delta_{xx} \Delta_{yy} \Delta_{xy}^2 + \frac{1}{48} \Delta_{xy}^4 \right) + \mathcal{O}(g^3) \right]. \quad (1.23)$$

The three order- g^2 vacuum diagrams are

$$\text{Diagram 1} + \text{Diagram 2} + \text{Diagram 3} \quad (1.24)$$

$S=128$ $S=16$ $S=48$

As at order g , all vacuum-disconnected terms cancel against the normalization by $Z[0]$. The connected result is therefore

$$G_{12} = -\hbar \Delta_{12} + \frac{g}{2} \hbar^2 \int d^d x \Delta_{1x} \Delta_{xx} \Delta_{x2} - g^2 \hbar^3 \int d^d x d^d y \left(\frac{1}{4} \Delta_{1x} \Delta_{2x} \Delta_{xy}^2 \Delta_{yy} + \frac{1}{4} \Delta_{1x} \Delta_{xx} \Delta_{xy} \Delta_{yy} \Delta_{y2} + \frac{1}{6} \Delta_{1x} \Delta_{xy}^3 \Delta_{y2} \right) + \mathcal{O}(g^3). \quad (1.25)$$

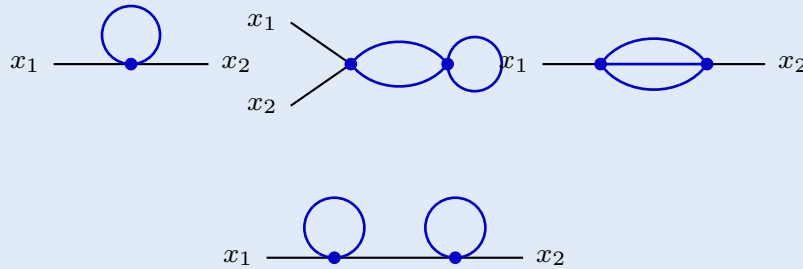
The order- g^2 terms are, in the same order as in the preceding equation,

$$\text{Diagram 1} + \text{Diagram 2} + \text{Diagram 3} \quad (1.26)$$

$S=4$ $S=4$ $S=6$

e) Mark the one-particle-irreducible subgraphs in the diagrams obtained in the previous part.

Solution.



f) Repeat the preceding two parts for the four-point correlation function, retaining all contributions through order g^2 .

Solution. Using the results from d), we have

$$G_{1234} = \hbar^4 \frac{\delta^4}{\delta_1 \delta_2 \delta_3 \delta_4} \exp \left[-\frac{1}{\hbar} \int d^4 x V \left(\hbar \frac{\delta}{\delta_x} \right) \right] \exp \left[-\frac{1}{\hbar} \frac{1}{2} J_x \Delta_{xy} J_y \right] \Big|_{J=0} \frac{Z_0[0]}{Z[0]}. \quad (1.27)$$

As before, all vacuum-disconnected terms cancel against the normalization by $Z[0]$. At order g^0 ,

we find

$$G_{1234}^{(0)} = \hbar^2 (\Delta_{12}\Delta_{34} + \Delta_{13}\Delta_{24} + \Delta_{14}\Delta_{23}). \quad (1.28)$$

The remaining disconnected terms are already fixed by the two-point result. For each of

$$(ij|kl) = (12|34), (13|24), (14|23), \quad (1.29)$$

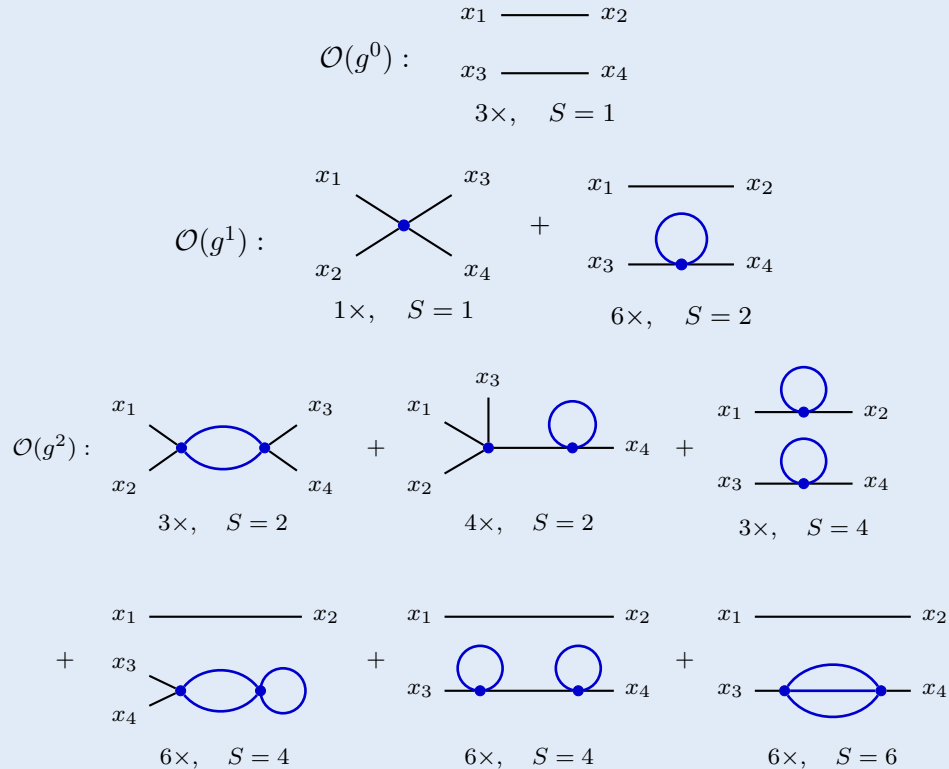
we obtain

$$\begin{aligned} G_{ij}G_{kl} &= \hbar^2 \Delta_{ij}\Delta_{kl} \\ &\quad - \frac{g}{2}\hbar^3 \int d^d x \left(\Delta_{ij}\Delta_{kx}\Delta_{xx}\Delta_{xl} + \Delta_{kl}\Delta_{ix}\Delta_{xx}\Delta_{xj} \right) \\ &\quad + g^2\hbar^4 \int d^d x d^d y \left\{ \Delta_{ij} \left(\frac{1}{4}\Delta_{kx}\Delta_{lx}\Delta_{xy}^2\Delta_{yy} + \frac{1}{4}\Delta_{kx}\Delta_{xx}\Delta_{xy}\Delta_{yy}\Delta_{yl} + \frac{1}{6}\Delta_{kx}\Delta_{xy}^3\Delta_{yl} \right) \right. \\ &\quad \left. + \Delta_{kl} \left(\frac{1}{4}\Delta_{ix}\Delta_{jx}\Delta_{xy}^2\Delta_{yy} + \frac{1}{4}\Delta_{ix}\Delta_{xx}\Delta_{xy}\Delta_{yy}\Delta_{yj} + \frac{1}{6}\Delta_{ix}\Delta_{xy}^3\Delta_{yj} \right) \right. \\ &\quad \left. + \frac{1}{4}\Delta_{ix}\Delta_{xx}\Delta_{xj}\Delta_{ky}\Delta_{yy}\Delta_{yl} \right\} + \mathcal{O}(g^3). \end{aligned} \quad (1.30)$$

The connected terms are

$$\begin{aligned} &G_{1234} - G_{12}G_{34} - G_{13}G_{24} - G_{14}G_{23} \\ &= -g\hbar^3 \int d^d x \Delta_{1x}\Delta_{2x}\Delta_{3x}\Delta_{4x} \\ &\quad + \frac{g^2}{2}\hbar^4 \int d^d x d^d y \left\{ \Delta_{xx}\Delta_{xy} \left(\Delta_{1x}\Delta_{2y}\Delta_{3y}\Delta_{4y} + \Delta_{2x}\Delta_{1y}\Delta_{3y}\Delta_{4y} \right. \right. \\ &\quad \left. \left. + \Delta_{3x}\Delta_{1y}\Delta_{2y}\Delta_{4y} + \Delta_{4x}\Delta_{1y}\Delta_{2y}\Delta_{3y} \right) \right. \\ &\quad \left. + \Delta_{xy}^2 \left(\Delta_{1x}\Delta_{2x}\Delta_{3y}\Delta_{4y} + \Delta_{1x}\Delta_{3x}\Delta_{2y}\Delta_{4y} + \Delta_{1x}\Delta_{4x}\Delta_{2y}\Delta_{3y} \right) \right\} + \mathcal{O}(g^3). \end{aligned} \quad (1.31)$$

These corresponds to the diagrams



g) Define

$$G_{\text{CC}}(x_1, x_2, x_3, x_4) = \hbar^3 \frac{\delta}{\delta J(x_1)} \cdots \frac{\delta}{\delta J(x_4)} \mathcal{W}[J] \Big|_{J=0}, \quad \mathcal{W}[J] = \hbar \log \mathcal{Z}[J]. \quad (1.32)$$

Show that this construction produces connected Feynman diagrams only.

Solution.

$$\begin{aligned} G_{\text{CC}}(x_1, x_2, x_3, x_4) &= \hbar^4 \delta_1 \delta_2 \delta_3 \left(\frac{\delta_4 \mathcal{Z}[J]}{\mathcal{Z}[J]} \right) \Big|_{J=0} \\ &= \hbar^4 \left(\frac{\delta_1 \delta_2 \delta_3 \delta_4 \mathcal{Z}[J]}{\mathcal{Z}} - \frac{\delta_1 \delta_2 \mathcal{Z}[J] \delta_3 \delta_4 \mathcal{Z}[J] + \delta_1 \delta_4 \mathcal{Z}[J] \delta_2 \delta_3 \mathcal{Z}[J] + \delta_1 \delta_3 \mathcal{Z}[J] \delta_2 \delta_4 \mathcal{Z}[J]}{\mathcal{Z}^2} \right) \Big|_{J=0} \\ &= G_{1234} - G_{12} G_{34} - G_{13} G_{24} - G_{14} G_{23} \\ &= (\text{connected4} - \text{pointdiagrams}), \end{aligned} \quad (1.33)$$

where we have used the fact that there are no tadpole diagrams for ϕ^4 theory.

2. Generating Functional for One-Particle-Irreducible Diagrams

Define the background field in the presence of J by

$$\varphi(x) = \frac{\delta \mathcal{W}[J]}{\delta J(x)} = \langle \phi(x) \rangle [J], \quad (2.1)$$

and introduce the effective action through the Legendre transform

$$\Gamma[\varphi] = \int d^4x J(x) \varphi(x) - \mathcal{W}[J]. \quad (2.2)$$

Recall that the connected generating functional has the expansion

$$\mathcal{W}[J] = \sum_{n=0}^{\infty} \frac{1}{n!} \left[\prod_{k=1}^n \int d^4x_k J(x_k) \right] G_{\text{CC}}(x_1, \dots, x_n), \quad (2.3)$$

where $G_{\text{CC}}(x_1, \dots, x_n)$ is the fully connected n -point function.

a) Derive the relation

$$J(x) = \frac{\delta \Gamma[\varphi]}{\delta \varphi(x)}. \quad (2.4)$$

Solution. Using chain rules, we have

$$\begin{aligned} \frac{\delta \Gamma[\varphi]}{\delta \varphi(x)} &= J(x) + \int d^4y \frac{\delta J(y)}{\delta \varphi(x)} \varphi(y) - \int d^4y \frac{\delta \mathcal{W}}{\delta J(y)} \frac{\delta J(y)}{\delta \varphi(x)} = J(x). \end{aligned} \quad (2.5)$$

b) Consider the Hessian

$$\frac{\delta^2 \Gamma}{\delta \varphi(x) \delta \varphi(y)}. \quad (2.6)$$

Show that its functional inverse is the connected two-point function,

$$G_{\text{CC}}(x_1, x_2) = \left[\frac{\delta^2 \Gamma}{\delta \varphi(x_1) \delta \varphi(x_2)} \right]^{-1}, \quad G_{\text{CC}}(x_1, x_2) = \langle \phi(x_1) \phi(x_2) \rangle_{\text{conn.}}. \quad (2.7)$$

You may use the functional chain rule

$$\frac{\delta}{\delta J(x)} = \int d^4y \frac{\delta \varphi(y)}{\delta J(x)} \frac{\delta}{\delta \varphi(y)}. \quad (2.8)$$

Solution.

$$\frac{\delta^2 \Gamma}{\delta \varphi(x_1) \delta \varphi(x_2)} = \frac{\delta J(x_2)}{\delta \varphi(x_1)} = \left(\frac{\delta^2 \mathcal{W}[J]}{\delta J(x_1) \delta J(x_2)} \right)^{-1} \implies G_{\text{CC}}(x_1, x_2) = \left[\frac{\delta^2 \Gamma}{\delta \varphi(x_1) \delta \varphi(x_2)} \right]^{-1}. \quad (2.9)$$

c) Fourier transform the connected two-point function according to

$$G_{\text{CC}}(x_1, x_2) = \int \frac{d^4 k}{(2\pi)^4} e^{ik \cdot (x_1 - x_2)} \tilde{G}(k). \quad (2.10)$$

Starting from its perturbative expansion, show that repeated 1PI self-energy insertions form a geometric series and hence give

$$\tilde{G}(k) = \frac{1}{k^2 + m^2 + M^2(k)}, \quad (2.11)$$

where $-M^2(k)$ denotes the sum of one-particle-irreducible insertions. The diagrammatic series is

$$\text{---} \bigcirc \text{---} = \text{---} + \text{---} \bigcirc \text{---} + \text{---} \bigcirc \bigcirc \text{---} + \dots$$

Solution. Note that at tree-level $\Gamma = S$, and therefore we can write

$$\Gamma_2(x, y) = G_0(x, y)^{-1} + \dots, \quad \Gamma_2(x, y) \equiv \frac{\delta \Gamma[\varphi]}{\delta \varphi(x) \delta \varphi(y)}. \quad (2.12)$$

In momentum space, this can be written as

$$\Gamma_2(k) = G_0(k)^{-1} + M^2(k) = \tilde{G}(k)^{-1}, \quad (2.13)$$

where $M^2(k)$ is the 1PI self-energy insertion. Using

$$\int d^d z \Gamma_2(x, z) G_{\text{CC}}(z, y) = \delta^d(x - y), \quad \Gamma_2(x, y) \equiv \frac{\delta \Gamma[\varphi]}{\delta \varphi(x) \delta \varphi(y)}. \quad (2.14)$$

This gives

$$\begin{aligned} \tilde{G}(k) &= \frac{1}{k^2 + m^2 + M^2(k)} \\ &= \frac{1}{k^2 + m^2} \sum_{n=0}^{\infty} \left(\frac{M^2(k)}{k^2 + m^2} \right)^n, \end{aligned} \quad (2.15)$$

which is exactly the geometric series.